

The Making of Vehicle Currencies

Liquidity Provision in DeFi/AMM Markets

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Nanyang Blockchain Conference 2026
NTU Singapore
21–22 August 2026



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Technologies

From the abstract to this talk

- The NBC abstract described three roles: quoting/invoicing, vehicle/intermediation, and reserve/settlement, with a flight-to-dominance story under stress.
- **This talk keeps only the vehicle/intermediation role, grounding it in liquidity provision instead of asset-pricing or flight dynamics.**
- **Dropped:** the quoting/invoicing and reserve/settlement roles, and any claim about flight-to-dominance as a pricing or attack phenomenon.
- What “flight-to-dominance” becomes here: liquidity depth and routing volume move together in the data (Proposition 1, with one contra-result left unresolved); architecture improves execution quality without changing which asset serves as the vehicle (Proposition 2).

The full three-role framing remains the long-run agenda. Today's talk covers the liquidity-provision leg only.

Outline

1. Motivation
2. The two propositions and their limits
3. Mechanism (borrowed theory, three analogies)
4. Evidence: formation (Proposition 1)
5. Evidence: architecture (Proposition 2)
6. Recap and contribution

Motivation: a vehicle-currency fact from FX

- In interdealer FX, roughly **13%** of USD-pair volume is vehicle-currency routing, a price-impact channel distinct from a spread-reducing one, rising to **25–38%** in major pairs (Somogyi 2026, quasi-experimental holiday design).
- That paper's own panel-regression mechanism test (its C1–C3 specifications) is explicitly non-causal, with an R^2 of roughly 18–20% dominated by fixed effects. We treat that mechanism test only as motivation; this talk does not rely on it as evidence.

The pivot

In FX, vehicle status is usually read off fundamental dollar demand. In DeFi, transaction-level liquidity-provision data lets us ask directly whether it instead traces to liquidity provision itself.

The two propositions and their limits

Proposition 1 (formation)

Vehicle-currency status is a liquidity-provision equilibrium: LP capital concentrates on a candidate's pools, deepens the pool, lowers the cost of indirect routing through it, and draws more LP capital, closing a self-reinforcing loop.

Proposition 2 (architecture)

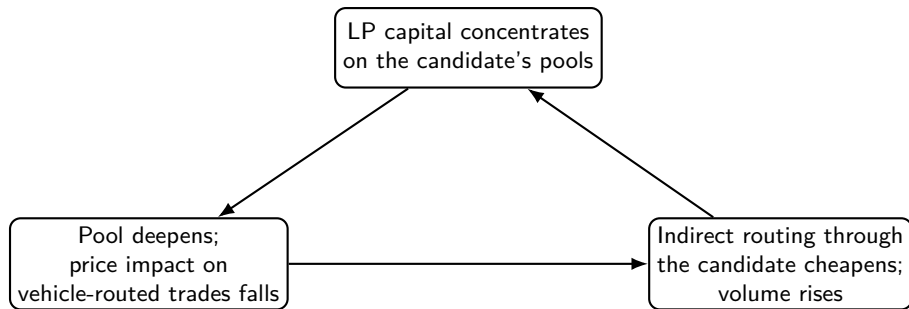
Architecture that cheapens LP repositioning makes vehicle status more elastic to stress.

Two constraints, named before the data

P1: no true exogenous shock is available in this sample. The one clean candidate (the UNI liquidity-mining launch, Sept. 2020) predates this repo's V3-only liquidity construction; what follows here is a lead-lag panel.

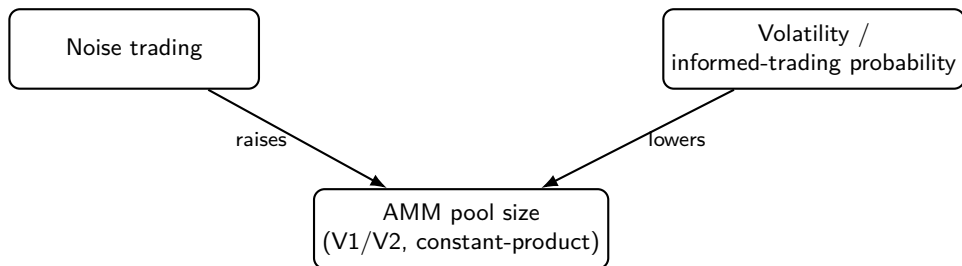
P2: a 2021 market-wide volatility episode sits on the same date used to test P2 (the V2→V3 launch), and this design cannot separate a true architecture effect from that episode in this sample.

Mechanism 1: the vehicle feedback loop (Krugman 1980, analogy)



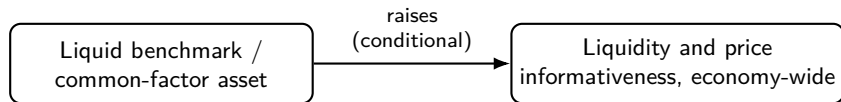
Motivating analogy only. Krugman's three-country transaction-cost mechanism, with an endogenized cost function. Scope here: at most three currencies, no persistence-after-dominance-fades claim, no empirical claim.

Mechanism 2: pool size and price impact (Lehár & Parlour 2024, analogy)



Cited only for their own result: under bounded conditions, AMM price impact can be lower and less volatile than a comparable limit-order book. Scope: their own 43-pair Binance comparison, constant-product $V1/V2$ only. This talk does not extend that result to a general AMM-versus-order-book claim or to $V3$.

Mechanism 3: benchmark liquidity (Yuan 2005, analogy)



Flagged as theory only, conditional on the specific assumption that market makers observe order flow across markets: no DeFi claim, no empirical claim, no assumption-free law.

Testing P1 with a lead-lag panel

- **Sample:** 5 candidates (WETH, USDC, USDT, DAI, WBTC), daily, 2021-05-05 to 2026-06-30, balanced ($1,883 \text{ dates} \times 5 \text{ tokens}$, $N = 9,415$).
- **Three series per token-day:** L = log candidate-linked liquidity; D = DirectCostAdvantage (execution-cost gap, direct vs. indirect route); S = indirect-route VehicleShare.
- **Design:** $\Delta_{\tau}(\text{outcome})_{t+\tau}$ on L_t, D_t, S_t , token + date fixed effects, $\tau \in \{1, 7, 14, 30\}$ days; cluster-by-date, Driscoll-Kraay, and month block-bootstrap inference.
- **What this shows:** L, D, S at t carry predictive content for each other's future changes beyond fixed effects. It cannot rule out a persistent common shock driving them together.
- **Independently re-derived** from scratch by two methods (sparse LSDV; from-scratch sandwich estimator); matches to 4 decimal places.

P1 headline: two legs support the loop, one is null, one runs backward

Relationship	Pred. sign	Evidence across the 4 horizons	Verdict
S on L	+	positive; $p < 0.001$ (DK) at all 4	Robust
L on S	+	positive; $p < 0.05$ (DK) at 3 of 4	Partial
S on D	–	correct sign; $p = 0.06$ – 0.24 , never sig.	Null
D on L	–	wrong sign at all 4; $p < 0.06$ (DK) at 2 of 4	Contra

“ Y on X ” = outcome $Y_{t+\tau}$ regressed on X_t . Full table in backup. Two bars this design does not clear: no exogenous shock/IV, and the fourth prediction runs backward (next slide).

P1's contra-result: one leg of the loop runs backward

P1 predicts deeper own liquidity should *lower* DirectCostAdvantage (D on L , $\beta < 0$). The pooled estimate is significantly positive at every horizon. It survives most, but not all, of the robustness battery:

- Recurs in *both* calm and high-volatility regimes.
- Recurs in 3 of 4 alternative depth/cost-measure constructions (the \$100k common-support variant is the one exception: it turns null instead of wrong-signed).
- Recurs in the pre-midpoint half of the sample; reverses (matches the predicted sign) post-midpoint.
- **Reverses** in the high/low baseline-depth split: both regimes match the predicted sign there (full detail in backup).

Bottom line

Two of four legs of the proposed loop show lead-lag support (S on L robust, L on S partial); one is null; one runs opposite to the theory in most but not all splits and stays unresolved.

Testing P2: V2 to V3 as an event study, reframed

- **Original ask:** does architecture that cheapens LP repositioning make vehicle status more elastic to stress? Tested via the V2→V3 concentrated-liquidity launch (2021-05-05).
- **Design note:** a cross-chain design was scoped and found not closeable (no multi-chain/L2 fetch path in this repo). Within-chain is the intended design, with chronological sample splits as the robustness check.
- **Sample:** 5 candidates, balanced pairs around the launch: ± 12 months (396 pairs, $N = 65,022$) and ± 24 months (507 pairs, $N = 97,484$).
- **Outcomes:** vehicle-candidate route share, vehicle HHI, direct-pool depth (level difference-in-differences), plus an elasticity test on $|\text{day-to-day change}|$ pre- vs. post-launch.

P2 headline: capital efficiency moves, topology does not

Window	Outcome	Effect	<i>p</i>
±12mo	Vehicle-candidate route share (pp)	0.94	0.141
±12mo	Vehicle HHI	0.014	0.246
±12mo	Direct-pool depth (ratio)	0.120	<0.001
±24mo	Vehicle-candidate route share (pp)	1.13	0.123
±24mo	Vehicle HHI	0.023	0.131
±24mo	Direct-pool depth (ratio)	0.120	<0.001

Direct-pool depth improves significantly post-V3 in both windows. Route share and vehicle HHI do not move. Architecture that improves execution quality does not automatically change which asset serves as the vehicle.

Why the elasticity claim is not testable here

Monthly mean |day-to-day Δ route share|,
model-free, from the raw panel:

Month	pp
2020-12	16.5
2021-01	21.4
2021-05 (launch)	22.2 (peak)
2021-06	16.8

Matches the well-documented Jan–May 2021
alt-season volatility episode, unrelated to the AMM
architecture change.

- Pre-launch placebo dates: large, significant “elasticity” jumps ($t = 4.4$ – 7.8).
- Post-launch placebos: mostly null.
- True launch window sits at the peak of the same cycle and comes back null.
- Joint pretrend Wald test: $\chi^2 = 49$ – 67 , $p < 0.001$ for every outcome, while the linear pretrend slope is insignificant ($p = 0.14$ – 0.99), consistent with a step-like calendar effect.

Not identifiable in this sample by this design.
Not claimed either way.

Recap and contribution

- Liquidity depth and routing volume move together in the direction P1 predicts (S on L robust across every check), with one contra-result left unresolved (D on L , wrong-signed across most, but not all, robustness splits).
- Architecture that improves execution quality (direct-pool depth up 0.12, $p < 0.001$) does not change which asset serves as the vehicle (route share and HHI both null); the P2 elasticity claim cannot be identified in this sample given a diagnosed 2021 confound.
- Capital efficiency and vehicle-currency status are separable margins.

A modest, partially-supported finding, not a closed causal loop.

Backup

Q&A coverage map | P1 backup: full headline table and *D-on-L* robustness detail
P2 backup: placebo dates, pretrend, volatility split, version construction
Design notes: data provenance, cost construction, why $V2 \rightarrow V3$ | References

What this appendix covers

- **Q&A coverage map:** the questions NTU's blockchain/CS-leaning audience is most likely to ask, each mapped to a specific slide below.
- **P1 backup:** the full lead-lag table ($4 \text{ relationships} \times 4 \text{ horizons}$) and the full *D-on-L* contra-result detail (every robustness split, plus the LINK falsification cells).
- **P2 backup:** the full placebo-date table, the joint-pretrend Wald detail, the volatility split, and the version-construction table.
- **Design notes:** data provenance and cost construction, why $V2 \rightarrow V3$ and not cross-chain, candidate-set and route construction.
- References, at full density.

Companion file to `slides/nbc_2026_core.tex`. Every number below is checked directly against a file under `output/nbc_pipeline/04_evidence/`.

Q&A coverage map (1/2): mechanics and data

Likely question	Backup slide
Does <i>D</i> (DirectCostAdvantage) include gas, or only fees/price impact?	"Data provenance and cost construction"
Where does the on-chain data come from, and how do you know an indexer/subgraph didn't drop or mis-attribute swaps?	"Data provenance and cost construction"
With only 5 tokens, is Driscoll-Kraay / date-clustering actually valid here?	"P1 backup: full headline table" (note)
How is an "indirect route" identified on-chain? Could a swap count as "through" a candidate just because the trader also held it?	"Candidate set, route construction, and the LINK placebo"
Could MEV/arbitrage bots mechanically generate the <i>S-on-L</i> result instead of liquidity providers doing so?	"Candidate set, route construction, and the LINK placebo"

Q&A coverage map (2/2): design choices and the contra-results

Likely question	Backup slide
Caparros-Chaudhary-Klein (CCK) is cited as the identification template and uses cross-chain gas variation. Why is P2 tested within one chain instead?	"Why V2→V3: no cross-chain design"
Why V2→V3 specifically? Why not V1, or a non-Uniswap AMM?	"Why V2→V3: no cross-chain design"
Is LINK really a fair placebo, given it's one unit with no fixed effects?	"P1 backup: LINK falsification detail"
Why this 5-token candidate set (WETH/USDC/USDT/DAI/WBTC), with no long-tail candidates?	"Candidate set, route construction, and the LINK placebo"
Is the V2→V3 null/confound really a design problem, or could it be a coding bug in the panel?	"P2 backup: joint pretrend detail" (note)

Per docs/nbc-2026-deck-pipeline.md: this is the one place NBC's actual audience profile is allowed to shape content, capped and confined to this appendix.

P1 backup: full headline table, all 4 relationships $\times$ 4 horizons

Relationship	τ	β	SE (cluster-date)	p (cl.-date)	SE (DK)	p (DK)	SE (bootstrap)
<i>S</i> on <i>L</i> (+)	1	0.0223	0.0015	<0.001	0.0015	<0.001	0.0024
<i>S</i> on <i>L</i> (+)	7	0.0261	0.0016	<0.001	0.0023	<0.001	0.0030
<i>S</i> on <i>L</i> (+)	14	0.0270	0.0017	<0.001	0.0030	<0.001	0.0032
<i>S</i> on <i>L</i> (+)	30	0.0270	0.0016	<0.001	0.0036	<0.001	0.0040
<i>L</i> on <i>S</i> (+)	1	0.0189	0.0091	0.039	0.0095	0.046	0.0094
<i>L</i> on <i>S</i> (+)	7	0.0377	0.0114	<0.001	0.0146	0.010	0.0130
<i>L</i> on <i>S</i> (+)	14	0.0483	0.0148	0.001	0.0247	0.051	0.0276
<i>L</i> on <i>S</i> (+)	30	0.0842	0.0154	<0.001	0.0344	0.014	0.0350
<i>S</i> on <i>D</i> (−)	1	−0.0060	0.0035	0.085	0.0036	0.095	0.0051
<i>S</i> on <i>D</i> (−)	7	−0.0057	0.0036	0.117	0.0049	0.239	0.0067
<i>S</i> on <i>D</i> (−)	14	−0.0122	0.0037	0.001	0.0065	0.060	0.0084
<i>S</i> on <i>D</i> (−)	30	−0.0129	0.0041	0.002	0.0084	0.123	0.0086
<i>D</i> on <i>L</i> (−)	1	0.0111	0.0058	0.054	0.0054	0.039	0.0080
<i>D</i> on <i>L</i> (−)	7	0.0162	0.0060	0.007	0.0085	0.057	0.0104
<i>D</i> on <i>L</i> (−)	14	0.0168	0.0064	0.008	0.0113	0.139	0.0131
<i>D</i> on <i>L</i> (−)	30	0.0244	0.0062	<0.001	0.0154	0.113	0.0156

$N = 9,265\text{--}9,410$; 1,853–1,882 date clusters, 5 tokens. Bold = wrong-signed vs. the predicted sign in parentheses. **Inference note (Q&A):** date-clustering (not token-clustering) is intentional: with only 5 tokens, clustering on the token dimension would give 5 clusters, which is invalid asymptotically. Clustering on date ($\sim 1,850\text{--}1,880$ clusters) plus Driscoll-Kraay (which runs on $T \rightarrow \infty$, N fixed) is the correct tool for $N = 5$; independently re-derived to 4 decimal places (Sec. 5, 05_verification.md). One nitpick from that check: the finite-sample multiplier uses $k = 4$ regressor d.f. without an FE adjustment, modestly understating SEs ($\sim 10\text{--}12\%$), so DK $p = 0.046\text{--}0.060$ above should not be read as robustly significant. Source: 04_evidence/p1/p1_headline_panel_results.md.

P1 backup: *D*-on-*L* detail — depth split (and a correction)

Depth regime	τ	β	p (DK)	Sign check
High baseline depth	1	-0.0245	<0.001	match
High baseline depth	7	-0.0244	0.008	match
High baseline depth	14	-0.0372	0.002	match
High baseline depth	30	-0.0236	0.084	match (n.s.)
Low baseline depth	1	-0.0449	<0.001	match
Low baseline depth	7	-0.0541	<0.001	match
Low baseline depth	14	-0.0617	<0.001	match
Low baseline depth	30	-0.0731	<0.001	match

Correction against the prose synthesis, checked against the primary table

The *D*-on-*L* wrong sign does **not** recur here in either regime: all 8 cells above match the predicted negative sign, most significantly. This is the one split where the contra-result reverses back to the theory's prediction, and it should not be read as a fifth confirmation of the wrong-signed pooled result. 05_verification.md's Sec. 3 prose describes this split as "recurs in the low-depth regime"; the primary table here, checked directly for this appendix, shows the opposite in both regimes. The wrong-signed cells the depth-split tally actually contains come from *S*-on-*D* and *L*-on-*S* instead of *D*-on-*L*; see next slides for where the contra-result really does recur.

Source: 04_evidence/p1/p1_robustness_depth_split.md, cross-checked against p1_robustness_depth_split.csv.

P1 backup: *D*-on-*L* detail — volatility split (recurs in both regimes)

Volatility regime	τ	β	p (DK)	Sign check
Calm	1	0.0071	0.409	wrong sign (n.s.)
Calm	7	0.0220	0.159	wrong sign (n.s.)
Calm	14	0.0232	0.282	wrong sign (n.s.)
Calm	30	0.0346	0.262	wrong sign (n.s.)
High	1	0.0117	0.106	wrong sign (n.s.)
High	7	0.0122	0.243	wrong sign (n.s.)
High	14	0.0137	0.316	wrong sign (n.s.)
High	30	0.0186	0.219	wrong sign (n.s.)

Every one of the 8 volatility-split cells for *D* on *L* is wrong-signed (positive, predicted negative); none reach DK significance individually, but the consistency of the sign across every horizon and both regimes is itself the finding this split contributes. Source:

04_evidence/p1/p1_robustness_volatility_split.csv (cross-checked against p1_robustness_battery_summary.md).

P1 backup: D -on- L detail — alternative measures

Variant	τ	β	p (DK)	Sign check
Alt. depth (lp_concentration)	1	0.0517	0.013	wrong sign
Alt. depth (lp_concentration)	7	0.0781	0.021	wrong sign
Alt. depth (lp_concentration)	14	0.0670	0.116	wrong sign (n.s.)
Alt. depth (lp_concentration)	30	0.0923	0.095	wrong sign (n.s.)
Winsorized-mean D	1	0.0133	<0.001	wrong sign
Winsorized-mean D	7	0.0195	<0.001	wrong sign
Winsorized-mean D	14	0.0222	<0.001	wrong sign
Winsorized-mean D	30	0.0286	<0.001	wrong sign
\$1k common-support D	1	0.0101	0.001	wrong sign
\$1k common-support D	7	0.0116	0.004	wrong sign
\$1k common-support D	14	0.0134	0.008	wrong sign
\$1k common-support D	30	0.0165	0.019	wrong sign
\$100k common-support D	1	-0.0075	0.164	match (n.s.)
\$100k common-support D	7	-0.0047	0.596	match (n.s.)
\$100k common-support D	14	-0.0065	0.558	match (n.s.)
\$100k common-support D	30	0.0035	0.802	wrong sign (n.s.)

Wrong-signed in 3 of 4 variants at every horizon (12 of 16 cells); the \$100k window is the one variant/horizon combination that turns null instead of wrong-signed, except at $\tau = 30$. Source: 04_evidence/p1/p1_robustness_altmeasure.csv.

P1 backup: *D-on-L* detail — period split and full tally

Period	τ	β	p (DK)	Sign check
Pre-midpoint	1	0.0192	0.087	wrong sign (n.s.)
Pre-midpoint	7	0.0263	0.094	wrong sign (n.s.)
Pre-midpoint	14	0.0285	0.110	wrong sign (n.s.)
Pre-midpoint	30	0.0327	0.183	wrong sign (n.s.)
Post-midpoint	1	-0.0618	0.002	match
Post-midpoint	7	-0.0717	0.014	match
Post-midpoint	14	-0.1026	0.008	match
Post-midpoint	30	-0.1376	0.001	match

Where the *D-on-L* contra-result actually recurs, checked directly

Pooled headline (wrong-signed, all 4 τ) → **volatility split: recurs in both regimes** → **alt-measure: recurs in 3 of 4 variants** → **period split: recurs pre-midpoint, reverses post-midpoint** → **depth split: reverses in both regimes** (previous slides). Net: this contra-result survives most but not all of the battery. A "nearly every split" characterization does not hold on its own; two of five checks (depth, post-midpoint period) actually flip back to the predicted sign.

Source: 04_evidence/p1/p1_robustness_period_split.csv.

P1 backup: LINK falsification detail

Equation	τ	β (LINK)	p (NW, LINK)	β (pooled)	Same sign?
S on L	1	0.0003	0.380	0.0223	yes
S on L	7	0.0001	0.911	0.0261	yes
S on L	14	-0.0008	0.183	0.0270	no
S on L	30	0.0001	0.898	0.0270	yes
L on S	1	-0.6536	0.152	0.0189	no
L on S	7	-2.4480	0.036	0.0377	no (wrong sign, significant)
L on S	14	-1.4347	0.363	0.0483	no
L on S	30	0.1612	0.900	0.0842	yes

Is LINK a fair placebo (Q&A)? Liquid, real V3 pools, a genuine occasional intermediate hop. It is not conventionally treated as a vehicle/bridge currency. Built by the identical L/S construction. But: single cross-sectional unit, no token+date FE possible, Newey-West HAC in place of Driscoll-Kraay – a strictly weaker, noisier estimator. Sign matches the pooled headline in only 4 of 8 cells; LINK's own coefficients are indistinguishable from zero in 7 of 8; one cell is wrong-signed and significant ($p = 0.036$). No placebo D was built: doing so would require rerunning the multi-year V2+V3 quote-simulation with LINK added, out of scope for this pass. This limit is disclosed in `build_link_placebo_panel.py`; it was not silently skipped. **Verdict: a weak, inconclusive placebo. It is not a clean pass either way.**

Source: 04_evidence/p1/p1_robustness_placebo_link.md.

P1 backup: full battery, remaining cells

TODO – formatting pass

The subsample sign-check tallies (depth 72 cells, volatility 72, alt-measure 144, period 72 – 360 cells total) are summarized on this appendix's earlier slides for the *D*-on-*L* relationship specifically. The remaining cells (the *S*-on-*L*, *L*-on-*S*, and *S*-on-*D* relationships across every split) are all directionally consistent with the headline (see

`p1_robustness_battery_summary.md`'s tallies: *S*-on-*L* is significant in all 32/32 depth×volatility×period×horizon cells and all 16/16 alt-measure×horizon cells). A full row-by-row table across all four relationships and all four split types is not reproduced here; pull directly from the CSVs below if a specific cell is requested live.

Sources:

`04_evidence/p1/p1_robustness_{depth_split,volatility_split,altmeasure,period_split}.csv`.

P2 backup: placebo dates, level DiD

Placebo date (regime)	Outcome	Effect	<i>p</i>
2020-10-01 (pre)	Route share (pp)	-1.7776	0.049
2020-10-01 (pre)	Vehicle HHI	-0.0224	0.049
2020-10-01 (pre)	Direct-pool depth	0.0554	0.003
2020-11-15 (pre)	Route share (pp)	-0.0901	0.873
2020-11-15 (pre)	Vehicle HHI	-0.0261	0.002
2020-11-15 (pre)	Direct-pool depth	0.0211	0.048
2021-01-01 (pre)	Route share (pp)	2.0903	<0.001
2021-01-01 (pre)	Vehicle HHI	0.0018	0.768
2021-01-01 (pre)	Direct-pool depth	0.0125	0.077
2022-05-05 (post, +1y)	Route share (pp)	-2.4767	<0.001
2022-05-05 (post, +1y)	Vehicle HHI	-0.0287	<0.001
2022-05-05 (post, +1y)	Direct-pool depth	-0.0255	0.043
2023-05-05 (post, +2y)	Route share (pp)	2.9108	<0.001
2023-05-05 (post, +2y)	Vehicle HHI	0.0150	0.061
2023-05-05 (post, +2y)	Direct-pool depth	-0.0067	0.301

Level DiD, ± 120 -day windows entirely inside one regime (never straddling 2021-05-05). Source: 04_evidence/p2/p2_robustness_battery_results.md, part (i).

P2 backup: placebo dates, elasticity test

Placebo date (regime)	Outcome (day-to-day Δ)	Effect	p
2020-10-01 (pre)	Route share (pp)	3.4318	<0.001
2020-10-01 (pre)	Vehicle HHI	0.0148	0.004
2020-10-01 (pre)	Direct-pool depth	-0.0192	<0.001
2020-11-15 (pre)	Route share (pp)	4.0151	<0.001
2020-11-15 (pre)	Vehicle HHI	0.0207	<0.001
2020-11-15 (pre)	Direct-pool depth	-0.0140	0.001
2021-01-01 (pre)	Route share (pp)	5.9643	<0.001
2021-01-01 (pre)	Vehicle HHI	0.0187	<0.001
2021-01-01 (pre)	Direct-pool depth	-0.0098	0.003
2021-05-05 (true launch)	Route share (pp)	-0.6810	0.621
2022-05-05 (post, +1y)	Route share (pp)	-0.6550	0.427
2022-05-05 (post, +1y)	Direct-pool depth	0.0219	<0.001
2023-05-05 (post, +2y)	Route share (pp)	1.1534	0.135
2023-05-05 (post, +2y)	Direct-pool depth	-0.0101	0.036

Every pre-launch placebo produces a large, significant “elasticity” jump; the true launch window does not. This is the falsification check failing at the pre-launch placebos and the pattern that motivates the pretrend test on the next slide. Source: 04_evidence/p2/p2_robustness_battery_results.md, part (i).

P2 backup: joint pretrend detail

Window	Outcome	Slope	Slope p	Joint χ^2 (12 df)	Joint p
$\pm 12\text{mo}$	Route share	0.0012	0.994	64.24	<0.001
$\pm 12\text{mo}$	Vehicle HHI	-0.0024	0.144	50.52	<0.001
$\pm 12\text{mo}$	Direct-pool depth	0.0176	<0.001	60.33	<0.001
$\pm 24\text{mo}$	Route share	0.0344	0.836	67.48	<0.001
$\pm 24\text{mo}$	Vehicle HHI	-0.0020	0.202	49.35	<0.001
$\pm 24\text{mo}$	Direct-pool depth	0.0184	<0.001	67.51	<0.001

A flat *linear* pretrend slope alongside a hugely significant *joint* test that all pre-launch month dummies are zero is a step-like, non-monotonic calendar effect. It is not a smooth trend toward the launch – exactly the signature the raw monthly $|\Delta \text{ route share}|$ series shows (16→21pp step-up, Dec 2020→Jan 2021, peaking at launch month, falling back by June 2021). **Bug or design problem (Q&A)?**

05_verification.md traced this with an independent code audit (the placebo estimator and headline estimator import and call the identical functions, so a latent bug would bias both in the same direction instead of selectively failing only the placebos) plus a raw-panel recomputation bypassing both scripts. Verdict: 85% confidence this is a genuine design/confound issue and not a coding error. Source:

04_evidence/p2/p2_robustness_battery_results.md, part (i), joint pretrend block.

P2 backup: volatility split, both windows

Window	Pre-V3 volatility	Outcome ($ \Delta $)	Effect	p
$\pm 12\text{mo}$	High	Route share (pp)	0.4028	0.823
$\pm 12\text{mo}$	Low	Route share (pp)	-1.1501	0.522
$\pm 12\text{mo}$	High	Vehicle HHI	-0.0030	0.778
$\pm 12\text{mo}$	High	Direct-pool depth	-0.0163	0.102
$\pm 12\text{mo}$	Low	Vehicle HHI	-0.0044	0.579
$\pm 12\text{mo}$	Low	Direct-pool depth	0.0047	0.137
$\pm 24\text{mo}$	High	Route share (pp)	1.6682	0.295
$\pm 24\text{mo}$	Low	Route share (pp)	-0.9029	0.670
$\pm 24\text{mo}$	High	Vehicle HHI	0.0010	0.906
$\pm 24\text{mo}$	High	Direct-pool depth	-0.0120	0.278
$\pm 24\text{mo}$	Low	Vehicle HHI	-0.0052	0.530
$\pm 24\text{mo}$	Low	Direct-pool depth	0.0143	0.002

Null in both volatility regimes for route share and vehicle HHI, every window. The one significant cell (low pre-V3 volatility, $\pm 24\text{mo}$, direct-pool depth, $p = 0.002$) is a capital-efficiency result. It is distinct from a vehicle-status/elasticity result, consistent with the headline's separation of the two margins.

Source: 04_evidence/p2/p2_robustness_battery_results.md, part (ii).

P2 backup: version construction, $\pm 12\text{mo}$

Version construction	Outcome	Effect	<i>p</i>
Best-across-versions	Direct-route availability (pp)	11.13	<0.001
Best-across-versions	Vehicle-route availability (pp)	25.99	<0.001
Best-across-versions	No-direct vehicle-route availability (pp)	-2.16	0.013
Best-across-versions	Direct-route quality (ratio)	0.139	<0.001
Best-across-versions	Direct cost advantage vs. vehicle route	-0.012	0.663
V2-only	Direct-route availability (pp)	-39.66	<0.001
V2-only	Vehicle-route availability (pp)	-28.18	<0.001
V2-only	No-direct vehicle-route availability (pp)	-2.29	<0.001
V2-only	Direct-route quality (ratio)	0.024	<0.001
V2-only	Direct cost advantage vs. vehicle route	-0.005	0.821
V3-only	Direct-route availability (pp)	50.79	<0.001
V3-only	Vehicle-route availability (pp)	28.77	<0.001
V3-only	No-direct vehicle-route availability (pp)	8.52	<0.001
V3-only	Direct-route quality (ratio)	-0.0002	0.321
V3-only	Direct cost advantage vs. vehicle route	0.0000	0.574

“Best-across-versions” is the default unfiltered panel (whichever version gave the best realized quote per leg); V2-only/V3-only restrict to legs whose realized best source was actually that version – a disclosed proxy for version availability. It is not a forced re-quote. Availability shifts mechanically with which version existed; cost-advantage effects are small and insignificant throughout. Source:

04_evidence/p2/p2_robustness_battery_results.md, part (iii).

P2 backup: version construction, $\pm 24\text{mo}$

Version construction		Outcome	Effect	<i>p</i>
Best-across-versions		Direct-route availability (pp)	11.51	<0.001
Best-across-versions		Vehicle-route availability (pp)	26.07	<0.001
Best-across-versions	No-direct vehicle-route availability (pp)		-2.31	0.012
Best-across-versions		Direct-route quality (ratio)	0.129	<0.001
Best-across-versions	Direct cost advantage vs. vehicle route		-0.002	0.932
V2-only		Direct-route availability (pp)	-43.89	<0.001
V2-only		Vehicle-route availability (pp)	-29.16	<0.001
V2-only	No-direct vehicle-route availability (pp)		-2.91	<0.001
V2-only		Direct-route quality (ratio)	0.017	0.003
V2-only	Direct cost advantage vs. vehicle route		0.001	0.970
V3-only		Direct-route availability (pp)	55.40	<0.001
V3-only		Vehicle-route availability (pp)	32.22	<0.001
V3-only	No-direct vehicle-route availability (pp)		7.95	<0.001
V3-only		Direct-route quality (ratio)	-0.0002	0.320
V3-only	Direct cost advantage vs. vehicle route		0.0000	0.873

Same pattern as $\pm 12\text{mo}$, both magnitude and significance. Source: 04_evidence/p2/p2_robustness_battery_results.md, part (iii).

P2 backup: fixed-window reproduction (robustness item iv)

The level-DiD and elasticity-test numbers under a fixed 12- and 24-month event window reproduce the headline exactly (Effect/SE/ t / p identical to the headline table in `p2_vehicle_status_elasticity_results.md`, to the reported decimal places). This confirms the headline windows are not a hand-picked variant that happens to look best: they are the same fixed-window construction the robustness battery independently re-derives. Source: `04_evidence/p2/p2_robustness_battery_results.md`, part (iv).

Data provenance and cost construction

- **Source:** swap/mint/burn/pool-day streams fetched from The Graph subgraphs (six sources: Uniswap V2/V3/V4, Curve, SushiSwap V3, Balancer, each pinned to a specific verified deployment ID) plus Dune for one venue (Fluid); stored as verbatim gzipped JSONL with per-day block-provenance sidecars (min/max block, head-block-at-fetch).
- **Reliability check:** a drift spot-audit re-fetches a random sample of days per source and diffs row IDs, counts, and block ranges against the stored archive. A drifting source is refetched in full; it is not patched cell by cell.
- **USD repricing:** subgraph-native amountUSD/amountInUSD fields are not trusted directly (a known derivedETH blow-up issue in the upstream derivation). Every swap leg is repriced from raw token amounts against a stablecoin-anchored per-day price table instead.
- **DirectCostAdvantage (D), does it include gas?** Yes. It is an all-in measure: registered pool fee, price impact from quote replay, and historical route-specific gas expenditure (route gas usage $\times$ day- t gas-token USD price), computed identically for the direct and indirect leg and differenced. Fee/impact/gas components are retained separately and sum back to the all-in figure within a numerical tolerance.
- **Candidate liquidity ($L_{k,t}$):** sum of Uniswap V3 pool TVL where the candidate is a pool side, restricted to pools with $0 < \text{TVL} \leq \$10\text{bn}$ (a spam-pool filter) and confirmed token contracts in the persisted V3 swap archive.

Sources: docs/data-reuse-plan.md; docs/research-questions-and-empirical-design.md (RQ1 Experiment A; variable registry entries for $C^{D,\text{gas}}$, $C^{I,\text{gas}}$); src/ddvc/variable_registry.py.

Why V2→V3: no cross-chain design

- Caparros, Chaudhary & Klein (CCK, 2024) is cited only for its own result (gas costs causally raise LP repositioning and concentration, across chains) as the identification template P2 was originally scoped to reuse.
- That specific reuse was scoped and found not closeable in this pass: this repo's data provenance is Ethereum-mainnet subgraph deployments only, with no multi-chain or L2 on-chain fetch path. Building one is real data-engineering work with its own new provenance and reliability burden. It is not a rerun of an existing pipeline.
- The V2→V3 concentrated-liquidity launch (2021-05-05) is the closest available within-chain shock to LP repositioning cost. Chronological sample splits ($\pm 12\text{mo}/\pm 24\text{mo}$, placebo dates, pretrend tests) substitute for the cross-chain contrast as this design's robustness axis. Java confirmed this substitution up front as the intended design.
- Cost of the substitution: it collapses CCK's actual source of variation (cross-chain gas-cost heterogeneity) into a single across-time architecture change, which is exactly why a market-wide volatility episode centered on the same calendar date becomes a live, undissolvable confound for the elasticity outcome (previous section). A cross-chain replication remains future work; it is not claimed as done here.
- Why V2/V3 and not V1: this repo's tick-level concentrated-liquidity construction is V3-only and populated from 2021-05-04 onward; extending it to V1/V2-equivalent liquidity is the same kind of non-trivial build work already declined for P1's exogenous-shock search (the September 2020 UNI liquidity-mining launch predates the V3-only $L_{k,t}$ construction for the identical reason).

Sources: 02_framings/framing_1.md Sec. 1 (P2 revision); docs/data-reuse-plan.md.

Candidate set, route construction, and the LINK placebo

- **Candidate set** $\mathcal{K} = \{\text{WETH, USDC, USDT, DAI, WBTC}\}$: prespecified in `src/ddvc/variable_registry.py`, the largest-TVL, most consistently bridging tokens across the sampled AMMs over the full window. It is not re-optimized on the outcome. A no-long-tail candidate set is a real scope limit, disclosed here: it has not been tested against smaller/altcoin candidates in this pass.
- **How “indirect route” is identified**: realized swap paths are reconstructed from the raw swap archive and classified by route class (direct single-hop vs. a coherent multi-hop path via a candidate). VehicleShare/BridgeShare counts a route as “through k ” only when k is an actual intermediate hop of a realized multi-leg swap. Merely holding the token alongside the trade does not count.
- **MEV/arbitrage bots**: this construction does not separately identify bot-driven multi-hop arbitrage flow from other trader flow. Bot flow is not excluded. The lead-lag design’s own stated limit covers this case: it establishes temporal precedence only, and does not rule out a common shock, such as a coordinated bot response to deepening liquidity, as an alternative to LP-driven feedback.
- **LINK as the falsification token**: chosen because it is liquid, has real V3 pools, and is a genuine occasional intermediate hop, but is not conventionally treated as a vehicle/bridge currency. See the dedicated LINK slide (P1 backup) for why the result reads as weak and inconclusive. It is not presented as a clean pass.

Sources: `src/ddvc/variable_registry.py`; `04_evidence/p1/build_link_placebo_panel.py` (docstring); `docs/research-questions-and-empirical-design.md` RQ1.

References

- Somogyi, F. (2026). FX vehicle-currency routing, quasi-experimental holiday design. Motivation only.
- Krugman, P. (1980). Three-country transaction-cost vehicle-selection mechanism. Motivating analogy for P1.
- Yuan, K. (2005). Benchmark/common-factor liquidity, conditional theory. Motivating analogy for P1.
- Lehár, A. and Parlour, C. (2024). AMM (V1/V2) pool size and price impact vs. a limit order book. Motivating analogy for P1.
- Klein, O., Kozhan, R., Viswanath-Natraj, G., and Wang, J. (2026). LP minting/burning and short-horizon return predictability. Motivates strategic LP behavior behind P1.
- Caparros, B., Chaudhary, S., and Klein, O. (2024). Gas costs, LP repositioning, and liquidity concentration. Identification template for P2 (cross-chain design not reused here; see “Why V2→V3: no cross-chain design”).
- Klein, O. and Song, Y. (2021); He, Z., Khorrami, P., and Song, Y. (2022). TradFi liquidity-commonality and intermediary-capacity analogies. Background only, no DeFi extension claimed.
- Barbon, A. and Rinaldo, A. (2024/Articles in Advance). CEX vs. DEX market quality, gas-fee causal effects. DeFi-native existence-proof comparator for venue/scope calibration. It is not a design template used above.

Full corpus manifest: `output/nbc_pipeline/00_manifest.md`. This appendix's own sources: `output/nbc_pipeline/04_evidence/{p1,p2}/`, `docs/research-questions-and-empirical-design.md`, `docs/data-reuse-plan.md`.